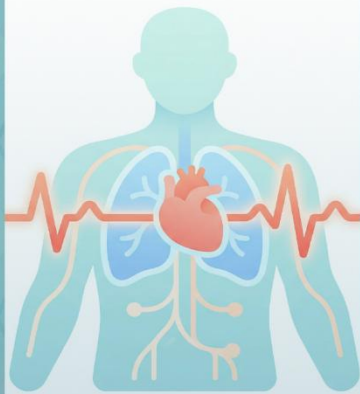




2026-27

CONCEPTS | CLINICALS | MCQS

PHYSIOLOGY

- Understand the Function.
Predict the Answer.
- Concept-based flowcharts &
mechanism clarity
- Graphs, cycles & most expected
MCQs



DR. KEYUR GANDHI

DR JITENDRA

PHYSIOLOGY

GENERAL PHYSIOLOGY

TYPES OF JUNCTIONS

Tight Junctions:

Tight junctions are seen between intestinal epithelial cells, PCT cells, and collecting duct cells towards their luminal side

Proteins which form tight junctions are:

1. Occludins
2. Claudins
3. JAMs (Junction Adhesion Molecules)

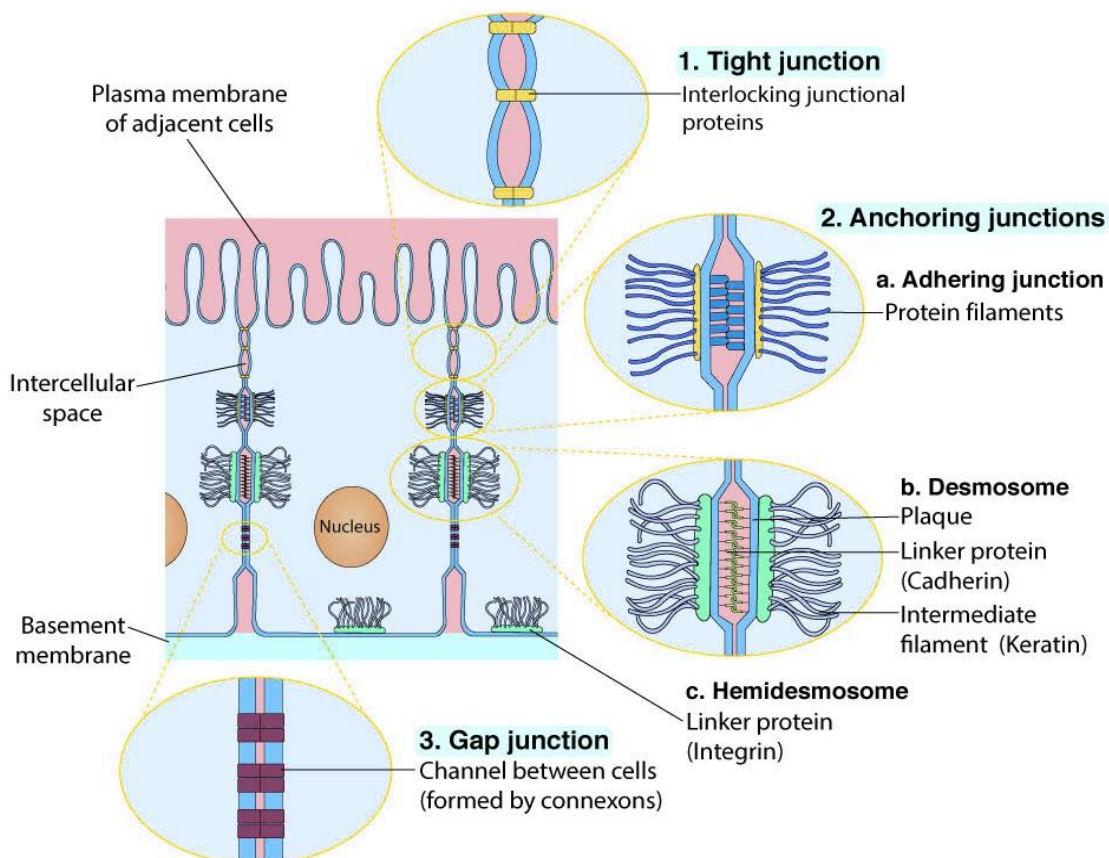
a. Zona occludens is a tight junction. eg. BBB

b. Zonula Adherence is made up of Adherens junction & Desmosomes.

Eg. E-cadherin-1 (seen in gastric carcinoma and lobular variant of ca. breast)

c. Hemidesmosomes: Attach the cell to the underlying basement membrane

TYPES OF CELL JUNCTIONS

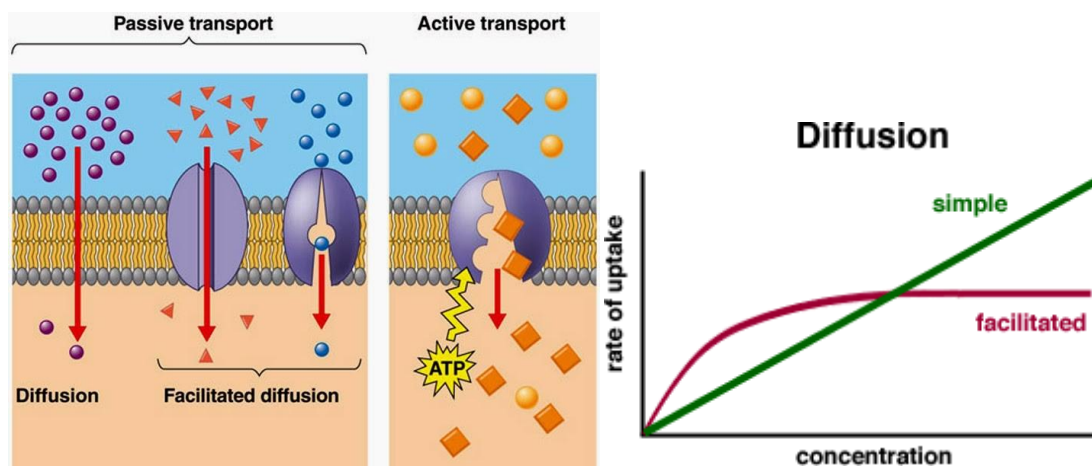


Gap junctions:

- Quick Transport (ICF of 1 cell → ICF of another)
- Syncytium
- Connexin → **Connexon**

Transport Across CM

Active transport	Passive transport
Against the gradient	Along the gradient
Needs energy is required	Does not need energy
Uphill movement	Downhill movement
1. Simple Diffusion 2. Facilitated diffusion 3. Non-Ionic diffusion 4. Osmosis	1. Primary active transport 2. Secondary active transport (a) Co-transport (Symport) (b) Counter- transport (Antiport)



Nerve Muscle

NERNST Potential/Equilibrium Potential for different IONS:

IONS	NERNST Potential
E_{K^+}	- 90 mV (- 95 mV)
E_{Cl^-}	- 70 mV
E_{Na^+}	+ 61 mV
$E_{Ca^{2+}}$	+ 125 mV

RMP of Different Cells:

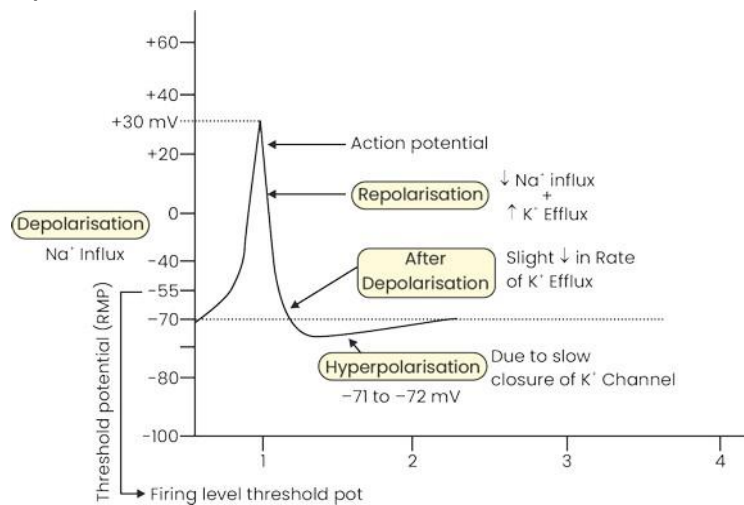
Cells	RMP
Most neurons	- 70 mV
Large motor Neurons	- 90 mV
Skeletal muscle Cell	- 90mV
Cardiac muscle Cell	- 90 mV
Pacemaker cell	- 50mV to - 60 mV
Smooth muscle cell	- 45 mV to - 65 mV

ACTION POTENTIAL

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Depolarisation – Influx of Na^+

Repolarization – K^+ Efflux



IMP 1 LINERS-

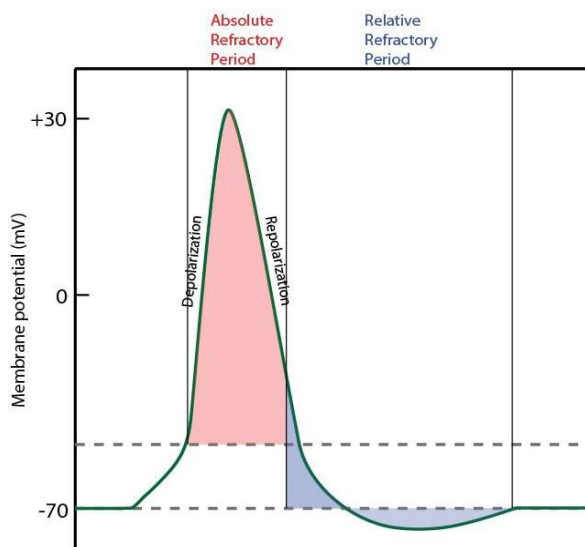
Resting membrane potential of most neurons is -70 mV

Resting Permeability through cell membrane: $\text{K}^+ > \text{Cl}^- > \text{Na}^+$

RMP of neuron is equal to equilibrium potential of Cl^-

RMP of myocardium is equal to equilibrium potential of K^+

At hyperpolarisation minimum combine transport of Na^+ and K^+



absolute refractory period.

A nerve fibre that cannot be stimulated despite applying stimuli of increasing intensity is in the absolute refractory period. In this state, sodium channels are closed, and potassium channels are open.

The absolute refractory period of a nerve is the period which another action potential cannot be triggered, no matter the strength of the stimulus applied. It starts from the point of the threshold potential to 1/3rd of repolarisation. Large number of Na channels closed.

SUSCEPTIBILITY TO NERVE FIBERS

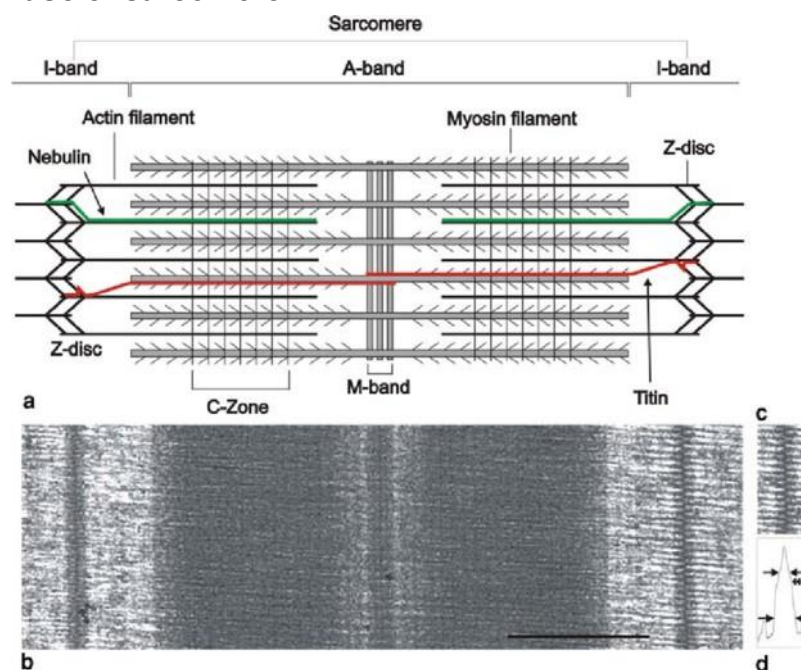
Sensitive to pressure- $A > B > C$

Sensitive to hypoxia- $B > A > C$

Paresthesias are usually seen when A-delta is involved.

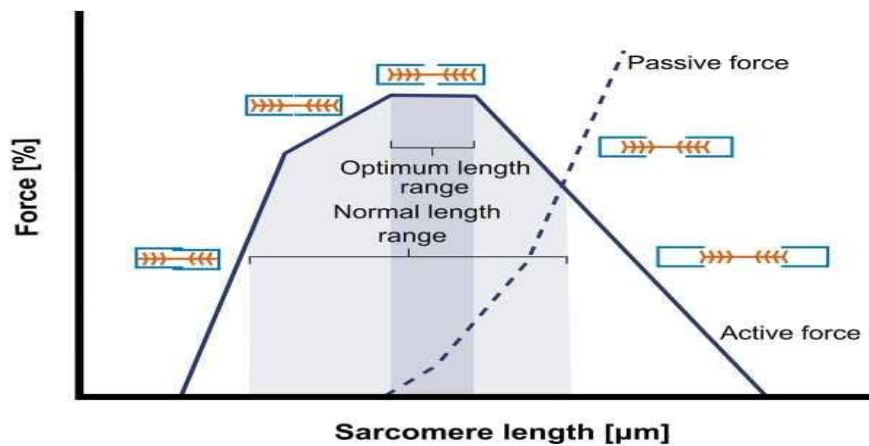
Number	Origin	Fiber Type
Ia	Muscle spindle, annulo-spiral ending	$A \alpha$
Ib	Golgi tendon organ	$A \alpha$
II	Muscle spindle, flower-spray ending; touch, pressure	$A \beta$
III	Pain and cold receptors; some touch receptors	$A \delta$
IV	Pain, temperature, and other receptors	Dorsal root C

Skeletal muscle- sarcomere



LENGTH TENSION RELATIONSHIP

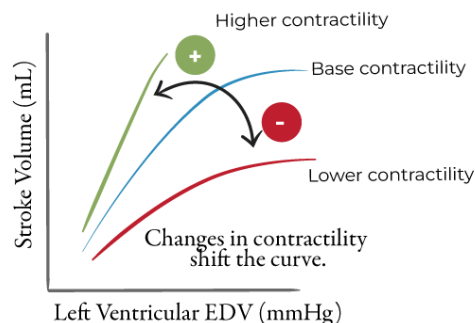
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***Frank Starling's Law:** It states that during isometric contractions, more the initial length, more the tension generated, but up to a physiological limit, beyond which further increase in initial length decreases the tension generated. Applicable to skeletal and cardiac muscle.

Frank-Starling Law

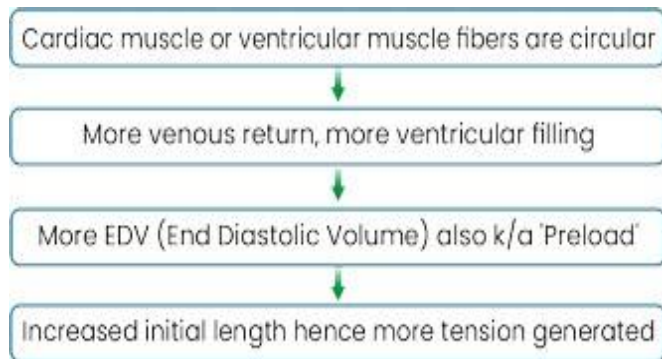
Shows that stroke volume changes as Preload/EDV does. The curve shifts right/left with changes in contractility.



INOTROPIC AGENTS

+	-
• Sympathetic NS • Adrenergic agonists • Cardiac glycosides (digoxin) • Angiotensin II • Low Na^+ • High Heart rate	• Parasympathetic NS (atria) • Beta blockers • Low Ca^{2+} (channel blockers) • Some antiarrhythmics • High Na^+ • Low Heart rate

Cardiac Muscle:



TACTILE RECEPTORS

Sensory receptors	function	Rapid adapting/slow adapting	Receptive field
Ruffini ending (subcutaneous)	Stretch of skin	Slow adapting	large
Meissner corpuscle (superficial)	Light touch of skin	Rapid adapting	small
Pacinian corpuscles	Pressure in skin	RA	large
Merkel disc (superficial)	Sustained in light touch	SA	small

BASAL GANGLIA DISEASE

Hypokinetic: Parkinsonism: substantia nigra (some hyperkinetic features also)

Hyperkinetic: Hemiballismus: subthalamic nucleus.

Huntington disease: Striatum (indirect pathway mainly)-involuntary movement similar to hemiballismus.

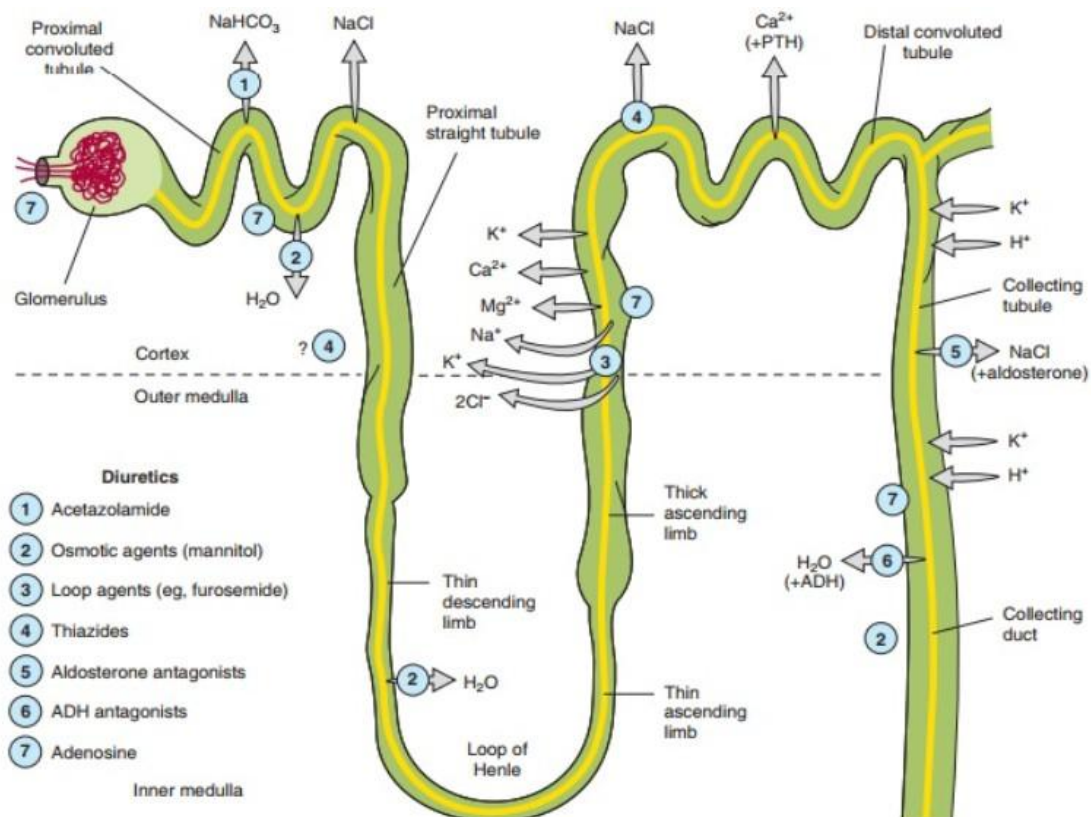
HYPOTHALAMUS

FUNCTIONS	HYPOTHALAMIC NUCLEUS
Neurosecretion	Supraoptic nucleus (ADH) Paraventricular nucleus (oxytocin)
hunger, thirst	Ventero-lateral hypothalamus
satiety	ventro-medial
Heat gain centre	Posterior hypothalamus

Heat loss centre	Anterior hypothalamus
Circadian rhythm	Suprachiasmatic nucleus

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KIDNEY



1 liners-

Large amount of water reabsorption from PCT by aquaporin-1

GFR= 180 l/day

Diluting segment of nephron- thick ascending limb of loop of Henle

Collecting duct is hormone regulated segment

Reabsorption of only water occurred in descending limb of loop of Henle

Diuretic which is anti-aldosterone → Spironolactone

Diuretic which acts on epithelial sodium channel → Amiloride

100 percent of glucose absorbed in PCT only

ADH controls water reabsorption in CD

ADH increases Aquaporin 2 on the luminal side of CD cells

ENaC-

- Gain of mutation – Liddle's Syndrome
- Loss of mutation – Pseudo Hypo Aldosteronism
- Aldosterone – stimulate – ENaC – ROMK – Na⁺ K⁺ ATPase – H⁺ K⁺ Exchanger

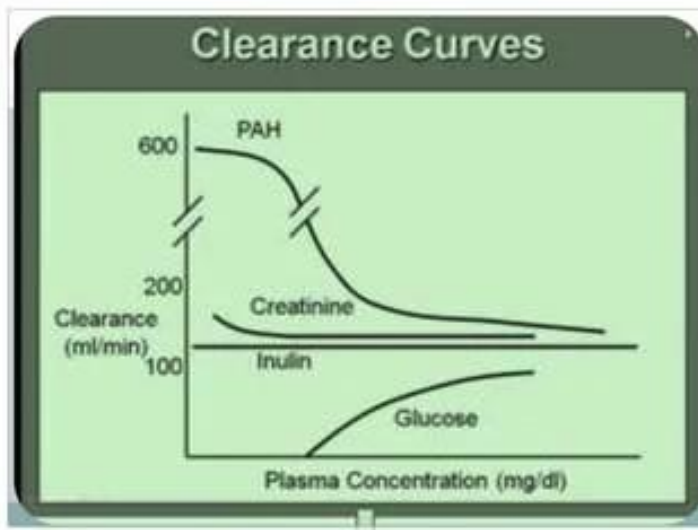
CLEARANCE FORMULA-

$$=VU/P$$

V- urine concentration

U- rate of urine flow

P- plasma concentration



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*STARLING'S EQUATION

$$\text{Net force} = (P\{c\} - P\{i\}) - (Pi\{c\} - Pi\{i\})$$

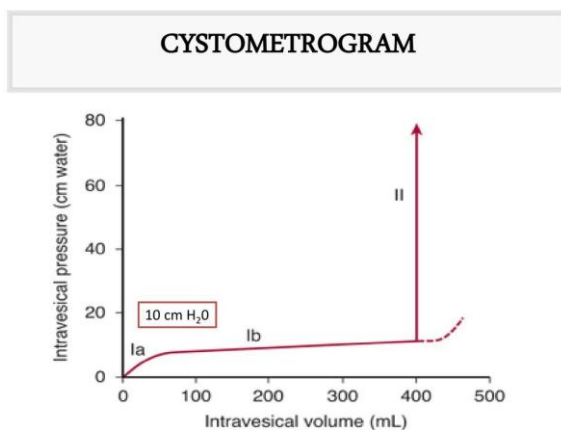
Where:

$P\{c\}$ = capillary hydrostatic pressure

$P\{i\}$ = interstitial hydrostatic pressure

$Pi\{i\}$ = interstitial oncotic (colloid osmotic) pressure

$Pi\{c\}$ = capillary oncotic (colloid osmotic) pressure



The minimal rise in pressure despite a substantial increase in volume, as depicted in graph (1b), is the high compliance of the bladder.

1a- upto 50 ml of feeling of bladder- pressure rises to 10cm H₂O

1b- flat part of curve

Laplace law

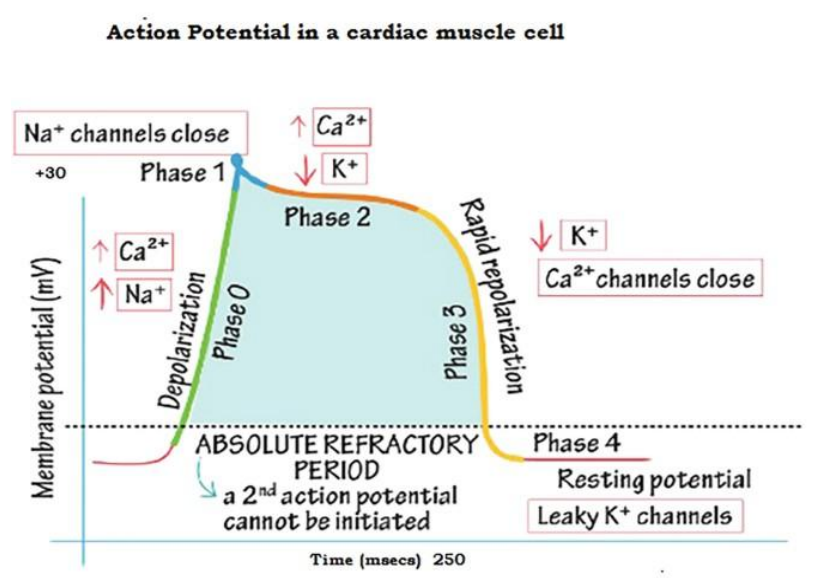
$$p = 2T (\text{tension}) / r$$

P- intravesical

400ml max volume of urine a patient can hold

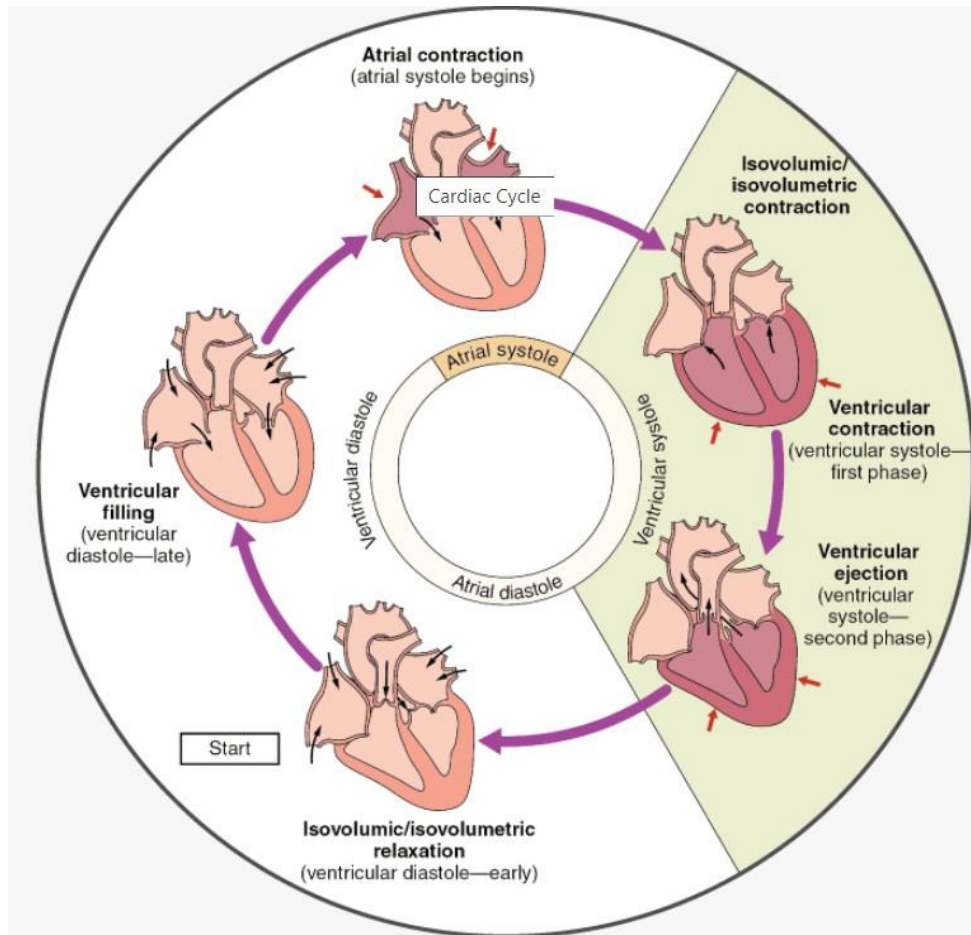
Afferent arterioles	RPF	GFR	FF
Constriction	↓↓	↓↓	N
Dilation	↑↑	↑↑	N
Efferent arteriole			
Constriction ^Q	↓↓	↑↑/↓↓	↑/↓
Dilation	↑↑	↓↓	↓

CVS



Presence of a plateau causes a widened absolute refractory period

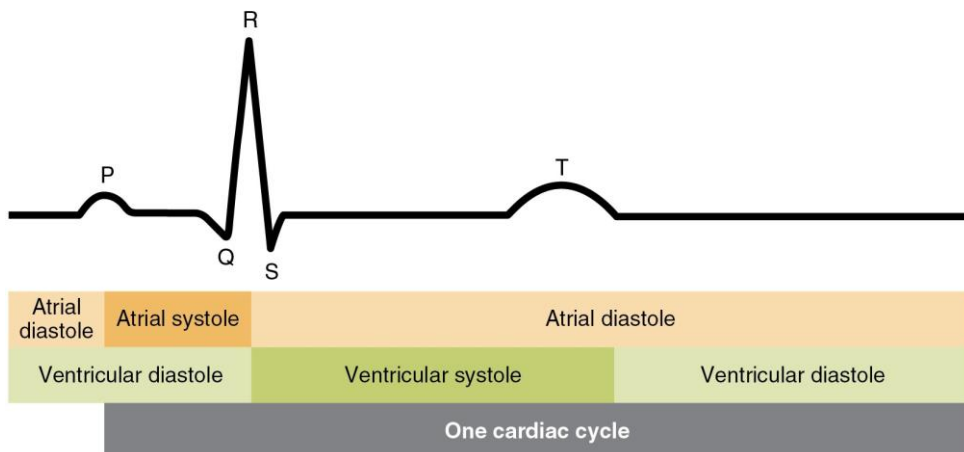
Due to the prolonged absolute refractory period (PARP) heart muscle cannot be tetanised



Cardiac cycle

All electrical and mechanical events which occur in the heart from one beat to the next are k/a cardiac cycle

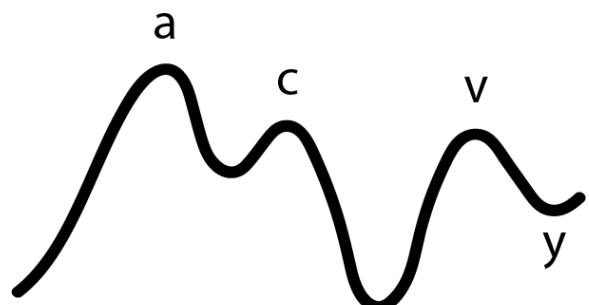
Normal cardiac cycle time = 0.8 second



JVP CHANGES WITH CARDIAC CYCLE

a = Atrial contraction

c = Due to bulging of closed tricuspid valve into RA during isovolumetric contraction



x = Due to the downward pull of the closed tricuspid valve into the right ventricle during the ejection phase

v = Due to the venous filling of the right atrium just before the opening of the tricuspid valve

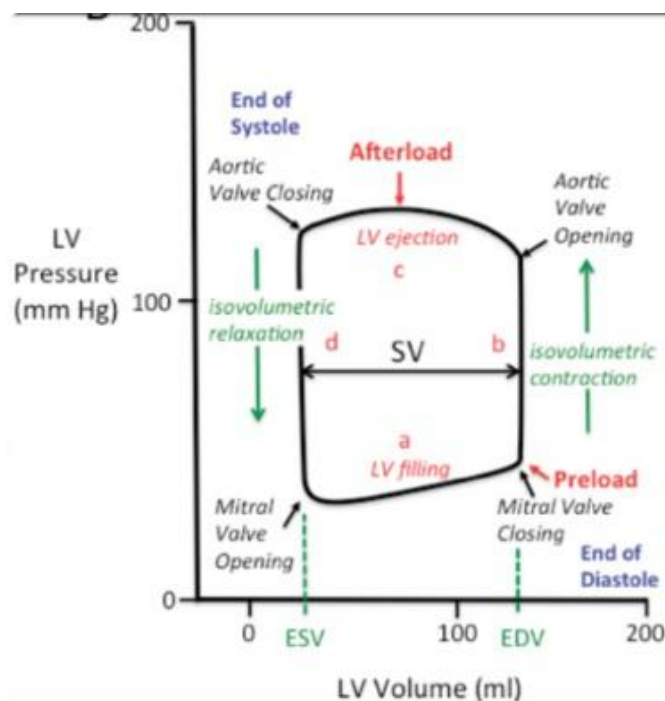
Y = Due to the flow of blood from the right atrium into the right ventricle during the first rapid filling phase

Cardiac output= LV output= RV output

Cardiac output= Heart rate * Stroke volume

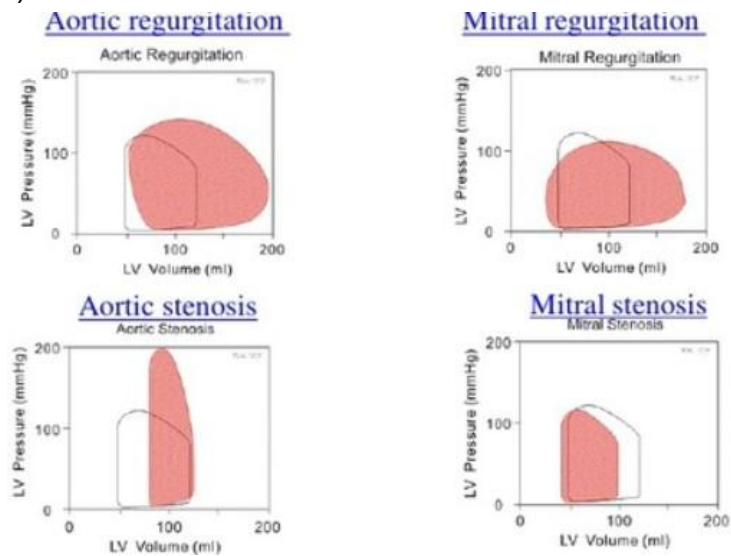
JVP ABNORMALITIES-

- **Absent a wave** - Atrial fibrillation, sinus tachycardia
- **Flutter waves** - Atrial flutter
- **Prominent a waves** - First-degree atrioventricular block
- **Large a waves** - Tricuspid stenosis, right atrial myxoma, pulmonary hypertension, pulmonic stenosis
- **Cannon a waves** - Atrioventricular dissociation, ventricular tachycardia
- **Absent x descent** - Tricuspid regurgitation
- **Prominent x descent** - Conditions causing enlarged a waves
- **Large cv waves** - Tricuspid regurgitation, constrictive pericarditis
- **Slow y descent** - Tricuspid stenosis, right atrial myxoma
- **Rapid y descent** - Constrictive pericarditis, severe right heart failure, tricuspid regurgitation, atrial septal defect
- **Absent y descent** - Cardiac tamponade



Q) opening of the aortic valve : image based

Q) MITRAL valve close : isovolumetric contraction



Aortic regurgitation- upward and right shift

Mitral regurgitation- right shift

Aortic stenosis- upward shift

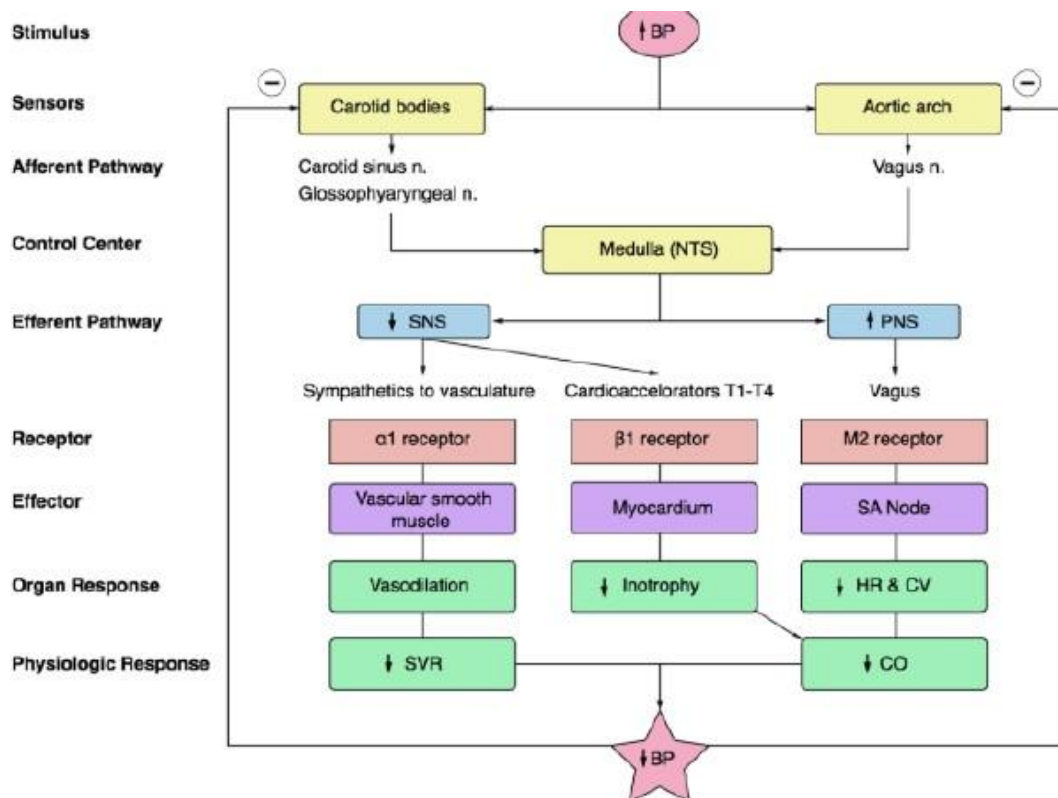
Mitral stenosis- downward and left shift

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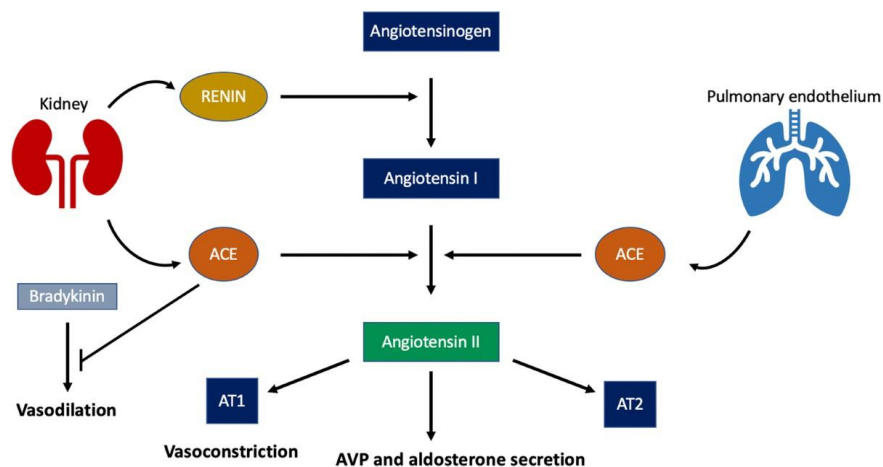
CVS Reflexes

1. Brainbridge Reflex - Rapid Infusion Increases VR → Stimulate Low Pressure Baroreceptors
– ↑ HR (Indirectly via medulla & Directly via SA node) – Tachycardia Reflex
2. Bezold-Jarisch Reflex - I.V. Irritant → Stimulate Ventricular Chemoreceptor – ↓HR, ↓BP, Apnea –Bradycardia Reflex
3. Cushing's Reflex - ↑ ICT = Compression Hypoxia → Stimulate VMC – ↑ BP, ↓ HR, Irregular Breathing –Bradycardia with Hypertension (fmge jan 26)

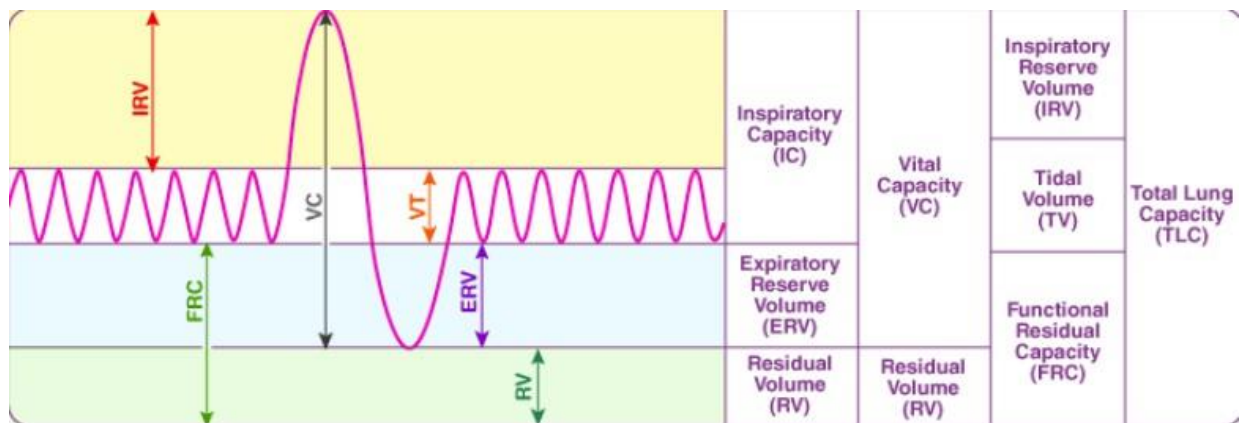
BARORECEPTOR REFLEX



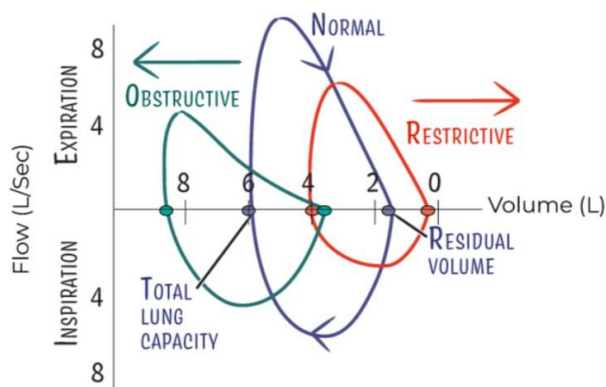
RAAS SYSTEM



SPIROMETRY-



Flow volume loop



obstructive	restrictive
Loop shift to left Volume are more than normal FEV1 decreases more than FVC Lower FEV1/FVC	Loop shift to right Volumes are lesser than normal FEV1 and FVC decrease in proportion Normal or elevated FEV1/FVC

Lung Compliance

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Compliance (C) = Distensibility

Compliance is inversely proportional to surface tension

Surface tension is inversely proportional to surfactant

Hence, compliance is directly proportional to surfactant

Specific lung compliance : is measured when pneumonectomy is done and want to check remaining lung's elasticity (fmge jan 2026)

SURFACTANTS

It reduces the surface tension

Surfactant is secreted by type II pneumocyte

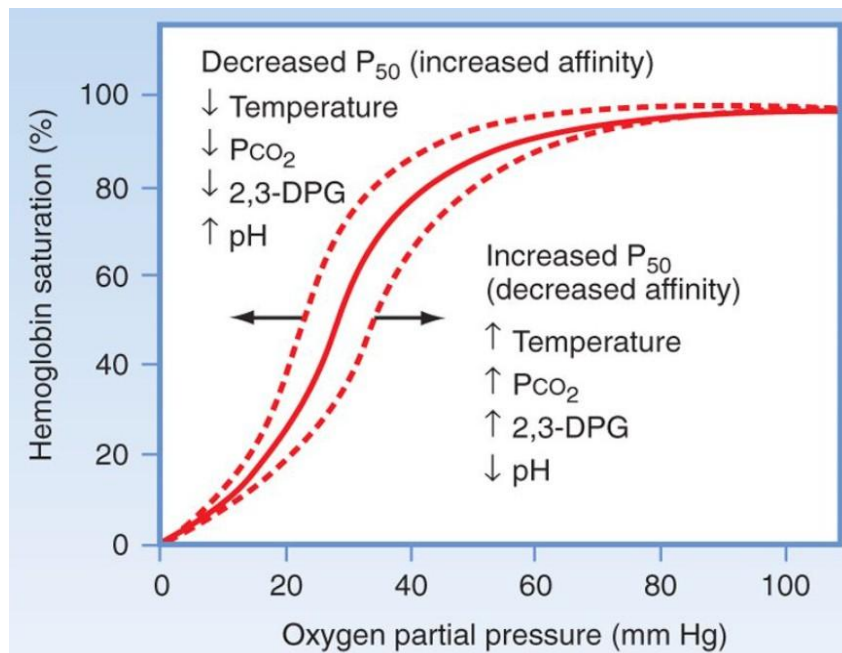
Chemical composition of surfactant – Dipalmitoyl phosphatidylcholine

Maximum Phospholipid - Phosphatidyl Phosphocholine (DPPC)/Lecithin

During expiration → decrease in alveolar size → increases the concentration of surfactant per unit area → causes a greater decrease in surface tension → prevents alveolar collapse (maintains alveolar stability)

OHDC (Oxygen Haemoglobin Dissociation curve)-

Graph of PO₂ and SO₂ correlation is sigmoid-shaped



Periodic Breathing:

- A. Cheyne Stokes Breathing → Hyperventilation followed by Apnea Gradual Increase and decrease. E.g. High Altitude, Sleep Apnea, CHF
- B. Biot's Breathing → Irregular rate and Rhythm E.g. Meningitis, Bulbar Polio, Uremia
- C. Kussmaul's Breathing / Air Hunger → Increased Rate and Depth

Regulation of respiration:

1. Role of medulla - Pacemaker of respiration in medulla is k/a pre-Botzinger complex

Medulla has a respiratory centre which has 2 parts- Dorsal respiratory group of neurons (DRG) & Ventral respiratory group of neurons (VRG)

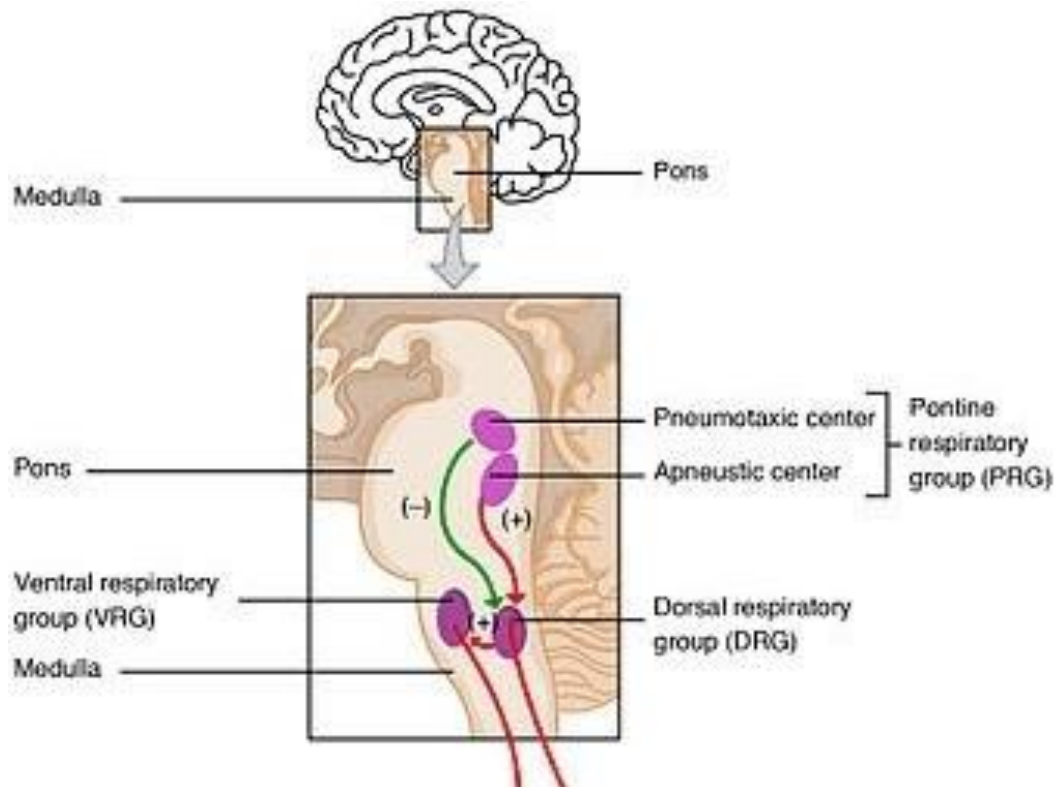
DRG- function in normal quiet respiration

VRG-Functions whenever the requirement of O₂ increases (e.g., Exercise)

2. Pontine Centres

Pneumotaxic Centre (Higher pons)- inhibit depth of inspiration

Apneustic Centre (Lower pons)- stimulate depth of inspiration

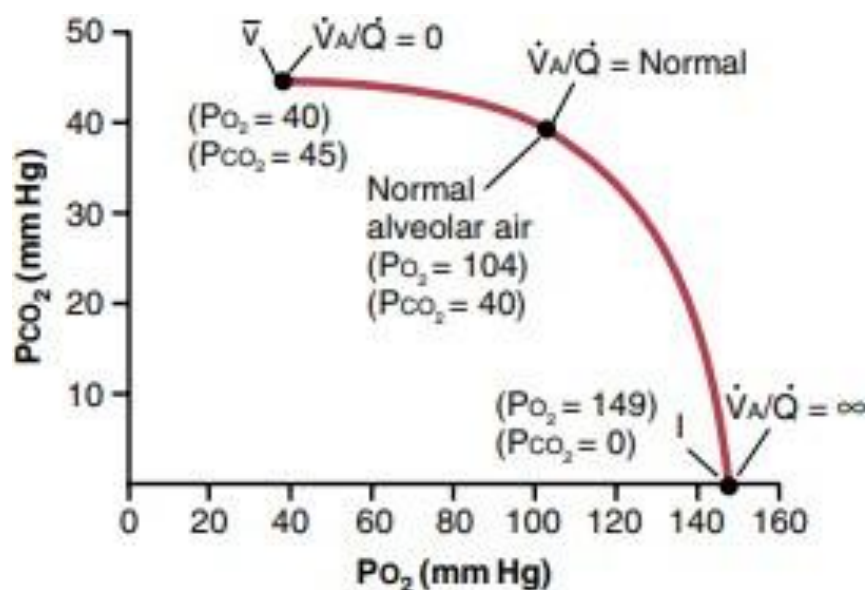


3. Ondine's curse/Central hypoventilation syndrome: Voluntary control of respiration is present but involuntary control is lost.

Site of the lesion is probably pre-Botzinger complex

VENTILATION PERFUSION RATIO

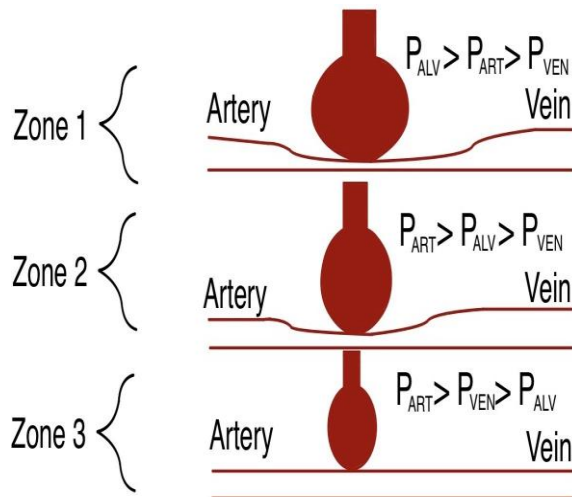
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V/Q ratio : zero : foreign body obstruction

V/Q ratio : infinite : pulmonary embolism

If perfusion to the alveolus is blocked by an embolus ($Q = 0$), gas exchange cannot occur. Although alveolar ventilation remains normal ($V > 0$), there is no blood flow to allow diffusion. Thus, the V/Q ratio becomes infinite (since the denominator, Q, is zero),



BREATHING ABNORMALITIES

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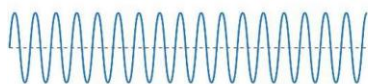
Eupnea (Normal breathing pattern and rate)
Causes: Balance diet and healthy life



Tachypnea (Fast and shallow breath)
Causes: Fever, anxiety, exercise, shock



Biot (Deep breath with periods of apnea)
Causes: Spinal meningitis, head injury



Kussmaul (Fast and deep breath)
Causes: Renal failure, diabetic ketoacidosis



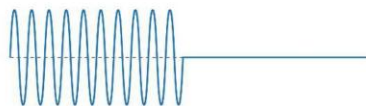
Bradypnea (Slow and shallow breath)
Causes: Sleep drugs, metabolic disorder



Sighing (Frequently interspersed sighs)
Causes: Anxiety, dyspnea, and dizziness

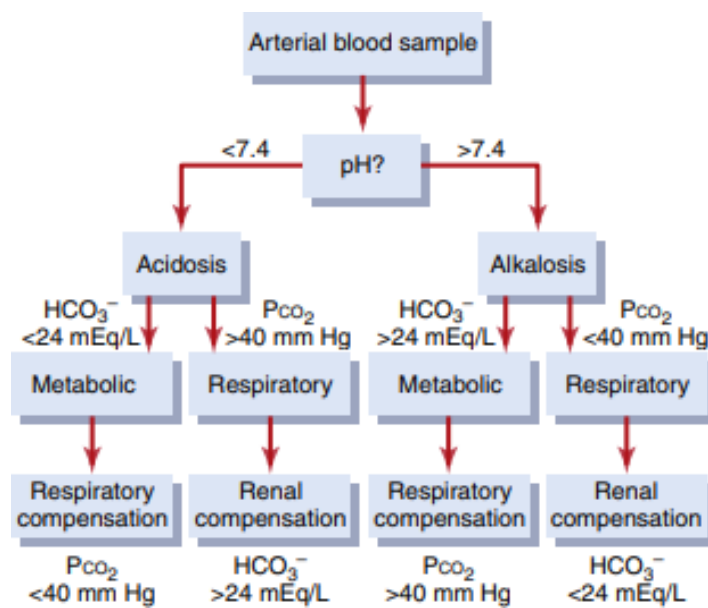


Cheyne-Stokes (Varying periods of breaths)
Causes: Brain tumors and injuries



Central Sleep Apnea (Breathing with apnea)
Causes: Heart and kidney failure

ACID BASE DISORDER



ENDOCRINE SYSTEM

Hormone which does not need second messenger

Growth hormone

Insulin

IGF-1

Prolactin

Hormone using cAMP as second messenger

ADH

Glucagon

PTH

Catecholamine (beta adrenergic receptors)

Hormone using cGMP as a second messenger

BNP

ANP

Prolactin- synthesis of milk

Oxytocin- expulsion of milk

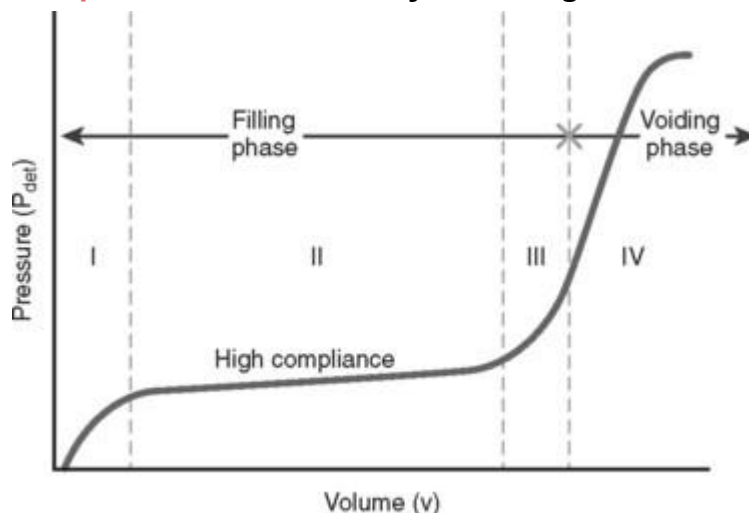
ANTIDIURETIC HORMONE

Mainly synthesised in 'Supra optic nucleus' & Paraventricular nuclei of the hypothalamus → Axoplasmic transport into the posterior pituitary → circulation

ADRENAL CORTICAL HORMONE- CORTISOL

Zones	Secretes	Under control of
Zona Glomerulosa	Aldosterone	RAAS
Zona Fasciculata	Glucocorticoids (mainly) + Androgens	ACTH
Zona Reticularis	Androgens (mainly) + Glucocorticoids	ACTH

- **Compliance of bladder : cystometrogram**



Minimum rise in pressure despite a substantial increase in volume (II) due high compliance of bladder .(FMGE 26)

GI TRACT

Pacemaker cells of GI tract: Interstitial cells of Cajal

Migrating Motor Complex

Waves of Contraction Produced By An Empty Stomach

Start from stomach & go till distal ileum

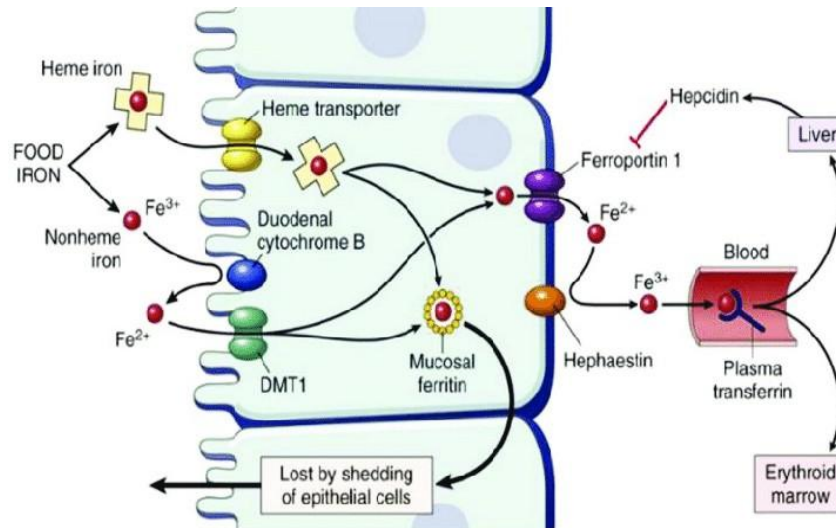
1MMC → 90-110 mins, Speed – 5cm/min

Stimulated by hormone motilin (Mo cells)- prokinetic drugs like Erythromycin also stimulate Mo cell

Iron absorption

Storage form in the enterocyte – Ferritin – Fe^{3+}

- Ferroportin – transfer Fe^{2+} from enterocyte to the blood
- Negative regulator of Ferroportin – **Hepcidin**
- Transferrin - help in the transportation of the ferric form



ABSORPTION

Jejunum - iron

Distal Ileum - B-12

Duodenum - iron

Hormonal control of GIT

HORMONES	SECRETED BY	FUNCTIONS
Gastrin	G cell (antrum)	-Increases HCl secretion
CCK	I cell (small intestine)	-Causes gallbladder contraction -Increases pancreatic enzyme secretion
GIP	K cell (Small intestine)	-Increases Insulin secretion
Ghrelin	D1/P cells (fundus of stomach)	-Orexigenic (Increases appetite)
Motilin	Mo Cell (Small intestine)	-Causes MMCS (Migrating Motor Complexes) Increases GUT motility

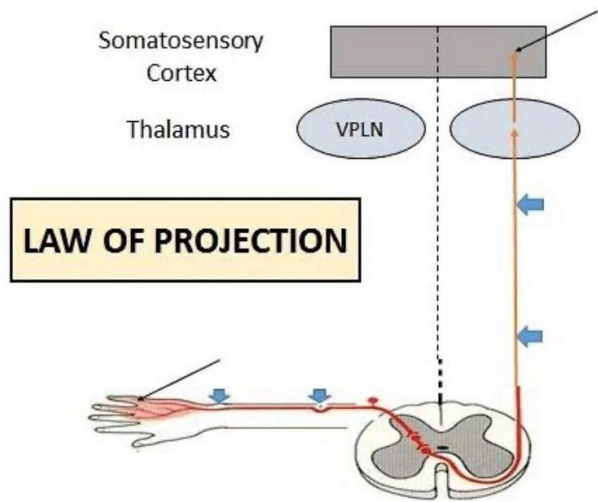
- Long term ppi use suppresses gastric acid secretion (reduce Hcl production)

Law of Projection- self

No matter where a particular sensory path is stimulated along its course to the cortex the conscious sensation produced is referred to the location of the receptor.

Eg. When the cortical receiving area for impulse for the left hand is stimulated, the patient reports sensation in the left hand not in the head.

Another example is phantom limb

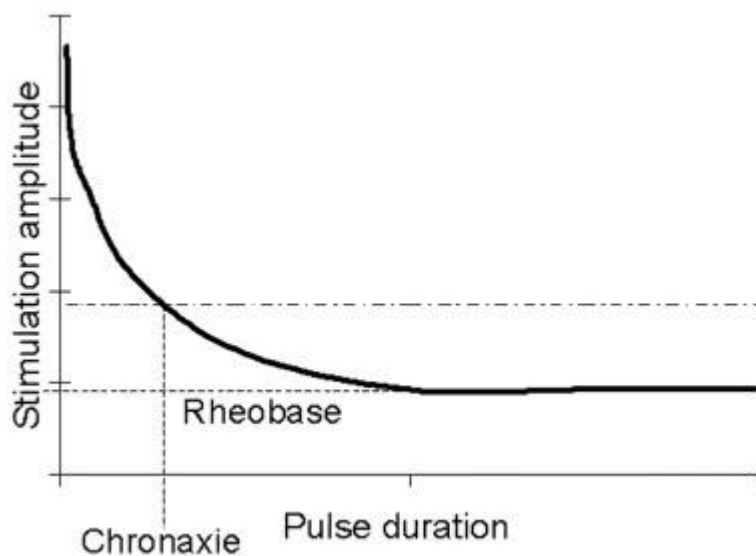


Isotonic vs isometric contraction

Isotonic contraction	Isometric contraction
Length of the muscle decreases	Muscle does not shorten
Tension in the muscle remains same	Tension in the muscles increases
Movement of joint occurs	No joint movement
Eg- lifting dumbbells	Eg- pushing against wall

HYPOXIA

Types	Hypoxic	Anemic	Stagnant	Histotoxic
Causes	High altitude Respiratory disease Defective diffusion	Anemia Hemorrhage Abnormal Hb	Cardiac failure Shock Circulatory failure	Cyanide poisoning Vit B deficiency
Arterial PO ₂	Low	Normal	Normal	Normal
Arterial O ₂ content	Low	Low	Normal	Normal
O ₂ carrying capacity	Normal	Reduced	Normal	Normal
A-V PO ₂ difference	Low	Normal	More than Normal	Nil
Oxygen therapy	100% oxygen administration is helpful	100% oxygen at high pressure is helpful	Limited value	No use



tissue strength duration curve

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TYPES OF DISTRIBUTIVE SHOCK

- SEPTIC SHOCK
- ANAPHYLACTIC SHOCK
- NEUROGENIC SHOCK

***NEUROGENIC SHOCK** : reduce tissue perfusion , loss of sympathetic tone ,

Everything will be low

Eg.. HR low

Bp low

Warm extremities

	HYPOVOEMIC	CARDIOGENIC	OBSTRUCTIVE	DISTRIBUTIVE
C0	LOW	LOW	LOW	HIGH
VR	HIGH	HIGH	HIGH	LOW
VENOUS PRESSURE	LOW	HIGH	HIGH	LOW
MIXED VENOUS SATURATION	LOW	LOW	LOW	HIGH
BASE DEFICIT	HIGH	HIGH	HIGH	HIGH

HIGH ALTITUDE :

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Decreased po_2 , causing cerebral vasodilation, leading to fluid leakage from the intravascular space. The low po_2 leads to hypoxic cerebral vasodilation.

Acute mountain sickness

MANAGED BY

Mild AMS : halting ascent, acetazolamide, and descent if needed.

Moderate AMS : descent, oxygen, acetazolamide or dexamethasone, and possibly hyperbaric therapy.